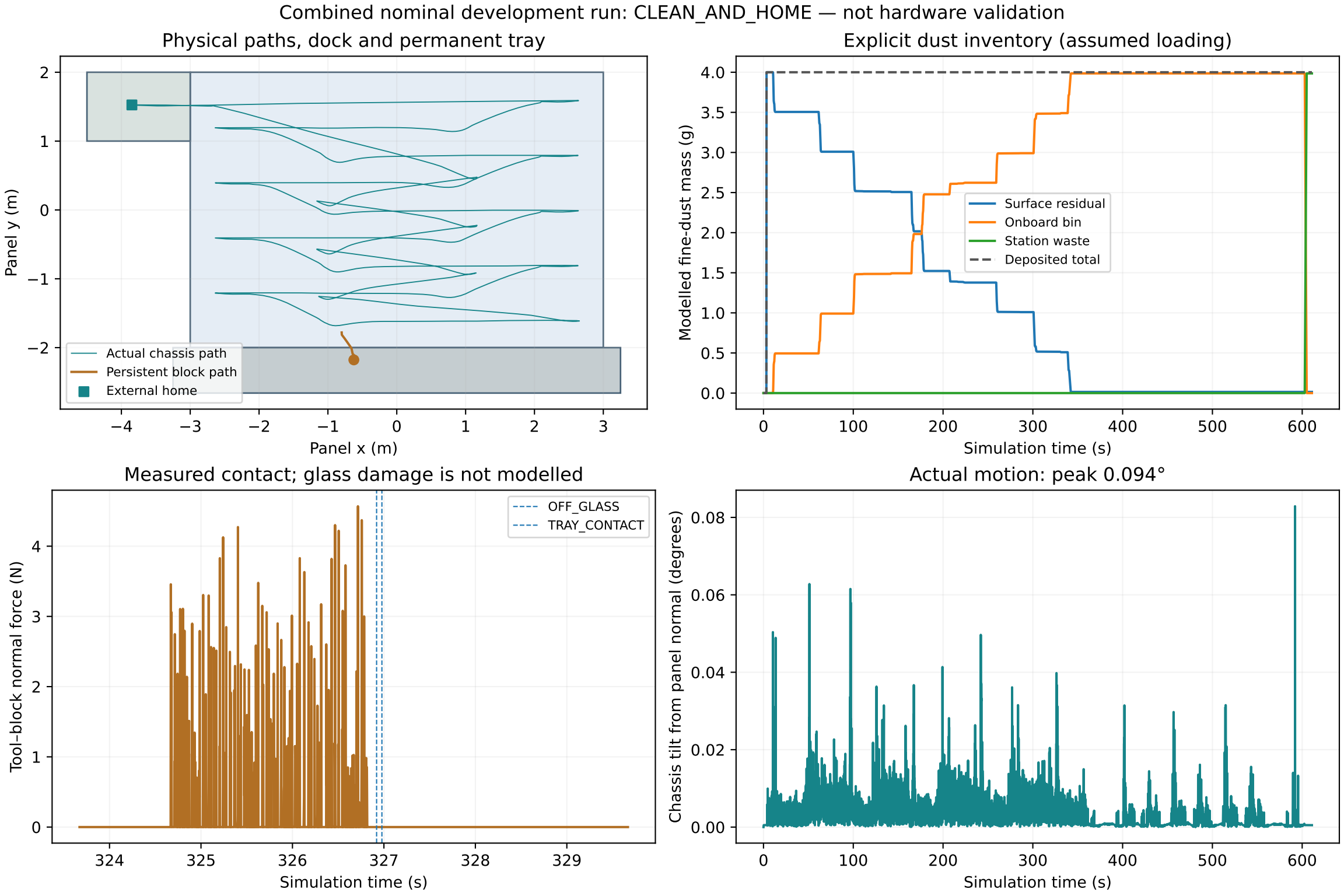

Combined nominal development run: CLEAN_AND_HOME — not hardware validation
Physical paths, dock and permanent tray
Panel y (m)
Panel x (m)
Actual chassis path
Persistent block path
External home
Explicit dust inventory (assumed loading)
Modelled fine-dust mass (g)
Simulation time (s)
Surface residual
Onboard bin
Station waste
Deposited total
Measured contact; glass damage is not modelled
Tool–block normal force (N)
Simulation time (s)
OFF_GLASS
TRAY_CONTACT
Actual motion: peak 0.094°
Chassis tilt from panel normal (degrees)
Simulation time (s)